1. O/N/23/21/Q1

Fig. 1.1 shows a toy water rocket.

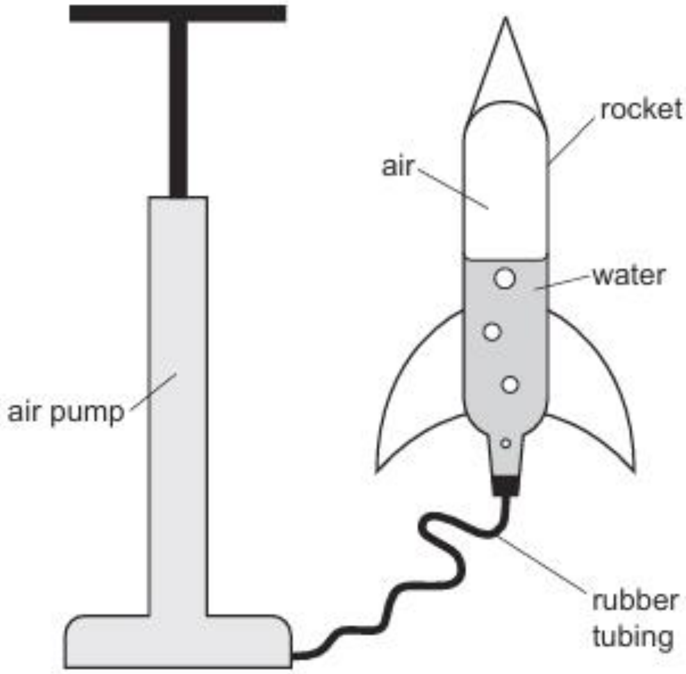

Fig. 1.1

The rocket is half-filled with water and connected by rubber tubing to a pump.

(a) Air is then pumped into the rocket so that the pressure of the air inside the rocket increases.

(i) Explain, in terms of particles, why the air in the rocket exerts a pressure on the walls of the rocket.

....

....

....

.... [3]

(ii) The temperature of the air in the rocket remains constant as air is pumped into the rocket.

Explain why increasing the number of air particles in the rocket increases the pressure of the air in the rocket.

....

.... [1]

(b) When the air pressure in the rocket is high, the rubber tubing is removed from the rocket.

The water in the rocket is expelled downwards.

Explain, in terms of momentum, why the rocket begins to accelerate upwards.

....

....

....

.... [3]

(c) As the water is expelled from the rocket, the upward acceleration of the rocket changes.

Suggest **two** reasons for the change in the acceleration of the rocket.

1 ....

....

2 ....

....

[2]

[Total: 9]

2. O/N/23/22/Q2

A spacecraft of mass 300 kg is moving in a straight line in space, at a speed of 8000 m / s.

(a) Calculate the momentum of the spacecraft.

momentum = ... kg m / s [2]

(b) The fuel on the spacecraft explodes and the spacecraft splits into two parts. The direction in which the parts move does not change.

(i) After the explosion, the speed of the front part increases to 9000 m / s. It has a mass of 150 kg.

Calculate the speed of the rear part after the explosion.

speed = ... m / s [3]

(ii) The total kinetic energy of the two parts after the explosion is greater than the original kinetic energy of the spacecraft.

State the energy transfer that occurs in the explosion.

....

.... [1]

(iii) The explosion lasts for 0.20 s.

Calculate the average force on the front part during this time.

force = ... N [3]

[Total: 9]

3. M/J/23/21/Q2

In a safety test, a car of mass 1100 kg travels at a speed of 10 m/s and collides with a stationary van of mass 3000 kg.

After the collision the car and the van move together with a velocity *v*.

Fig. 2.1 shows the car and van before and after the collision.

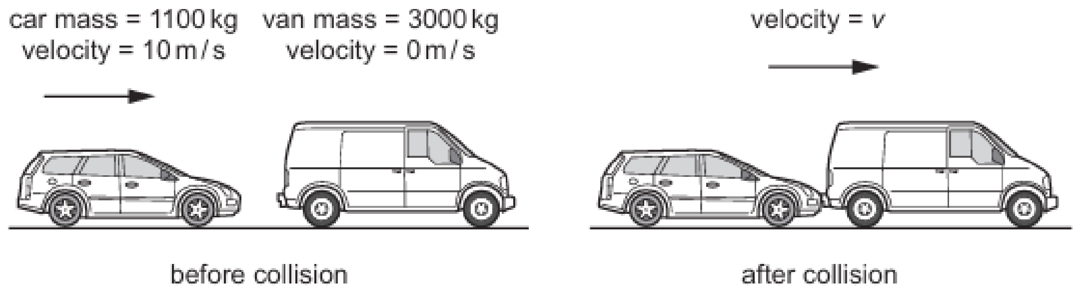

Fig. 2.1

The total momentum of the car and van is conserved during the collision.

(a) **(i)** Define 'momentum'.

....

.... [1]

(ii) State the unit of momentum.

.... [1]

(b) Calculate the velocity *v* of the car and van after the collision.

v = ... m/s [2]

(c) **(i)** Calculate the total kinetic energy of the car and van after the collision.

kinetic energy = ... J [2]

(ii) State the transfer of energy that occurs in the collision.

....

.... [1]

[Total: 7]

4. M/J23/22/Q1

Fig. 1.1 shows the speed–time graph for a car travelling on a straight horizontal road.

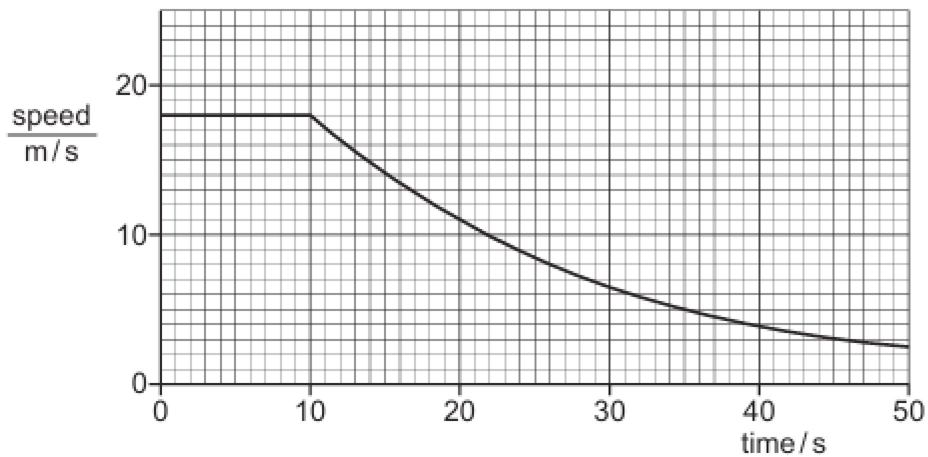

Fig. 1.1

(a) Describe the motion of the car shown in Fig. 1.1.

....

....

....

.... [2]

(b) At time t = 10 s the engine of the car is switched off. The brakes are not applied.

(i) Name **two** forces that act on the car to cause the change in motion after t = 10 s.

1 ....

2 ....

[1]

(ii) Suggest why Fig. 1.1 is a curve after t = 10 s.

....

....

.... [1]

(c) Between t = 10 s and t = 20 s the speed of the car changes from 18 m/s to 11 m/s.

The mass of the car is 1200 kg.

(i) Calculate the change in momentum of the car in this time.

Give the unit of your answer

momentum change = ... unit .... [2]

(ii) Calculate the average resultant force exerted on the car during this time.

average resultant force = ... N [2]

[Total: 8]